The shortest way

Optimization · Level 1 · Lesson 2



In this lesson you will

- Add up road lengths to find how long a way is.
- Use the closest-first trick to find the shortest way.
- Find a new way when a road is closed.

Key ideas

- Don’t guess by looking: add it up
- Settle the smallest number first
- Bridge closed: 14 km instead of 13
- Who walks the farthest?

Activity: Map detectives

1. Guess (3 min): Which way looks shortest?
2. Add (5 min): Add up two ways on the map.
3. Trick (8 min): Settle villages, smallest first.
4. Flood (4 min): Close the bridge and try again.

Copy the map from the lesson onto paper. No device needed.

Let’s discuss

1. Why did the straight-looking way lose?
2. What changed when the bridge flooded?
3. Who walks the farthest to your school?

Exit ticket

- The shortest way is ___ km. With the bridge closed, it is ___ km.